



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
22-08-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S & I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

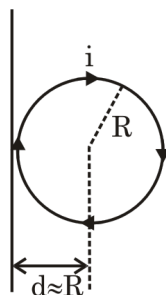
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

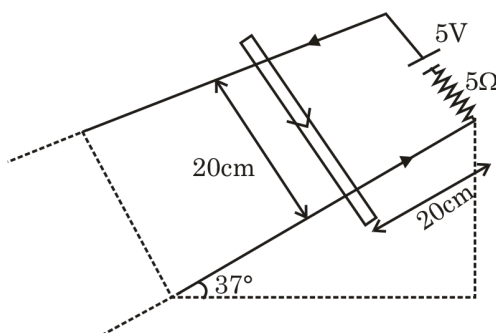
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A long conducting wire is placed in plane of a ring of radius R at distance $d \approx R$ from centre. Current in ring is increasing at constant rate α . Then induced emf in wire :

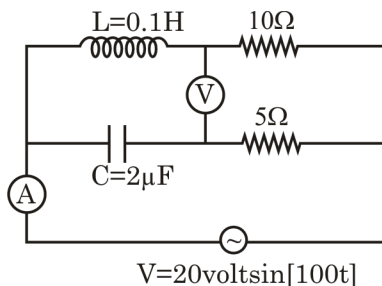


- (A) $\mu_0 R \alpha$ (B) $\frac{\mu_0 R \alpha}{2}$ (C) $\pi \mu_0 R \alpha$ (D) $\frac{\mu_0 R \alpha}{\pi}$
2. A conducting wire of (mass 0.1 kg) is placed on conducting rails inclined at angle 37° from horizontal. Ends of rails are connected by 5V, 5Ω battery when magnetic field $B = 2T$ is switch on in vertically upward direction then acceleration of wire just after magnetic field is switch on {self inductance of current loop is 0.2 H initially, friction coefficient between wire and rails is $3/4$ }.

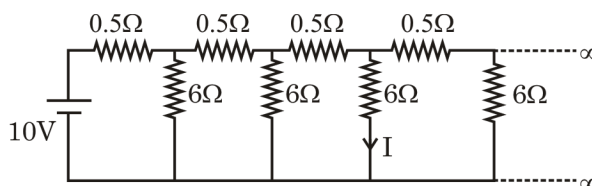


- (A) 0 (B) 2.95 m/s^2 (C) 1.25 m/s^2 (D) $4/15 \text{ m/s}^2$

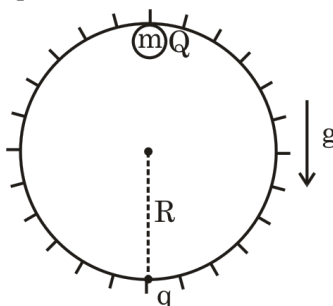
3. Reading of voltmeter and ammeter is :-



- (A) $10\sqrt{2}$ volt, $\sqrt{10}$ A
 (B) 10 volt, $\sqrt{5}$ A
 (C) $10\sqrt{2}$ volt, $\sqrt{5}$ A
 (D) 10 volt, $\sqrt{10}$ A
4. Value of current I is :-



- (A) 5A
 (B) $\left(\frac{15}{4}\right)$ A
 (C) $\left(\frac{45}{64}\right)$ A
 (D) None
5. A small ball of mass m, charge Q is to be held in stable equilibrium for small angular displacement as shown in figure. Find minimum charge q that should be fixed below ball for this :-



- (A) $\frac{16\pi\epsilon_0 mgR^2}{Q}$
 (B) $\frac{32\pi\epsilon_0 mgR^2}{Q}$
 (C) $\frac{8\pi\epsilon_0 mgR^2}{3Q}$
 (D) $\frac{64\pi\epsilon_0 mgR^2}{Q}$
6. An asteroid of mass M exploded into a spherical homogeneous cloud [spherical shell] in free space. Due to energy received by the explosion, the cloud expands and the expansion is spherically symmetric. At an instant, when radius of the cloud is R_0 , all of its particles on the surface are observed receding radially away from the centre of the cloud with a velocity v_0 . What will be the radius of the cloud, when its expansion ceases?

- (A) $\frac{GM R_0}{GM - v_0^2 R_0}$
 (B) $\frac{GM}{v_0^2}$
 (C) $\frac{2GM R_0}{2GM - v_0^2 R_0}$
 (D) $\frac{GM R_0}{2GM - v_0^2 R_0}$

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

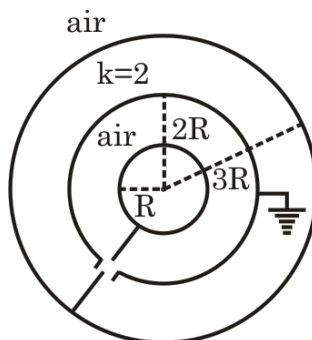
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

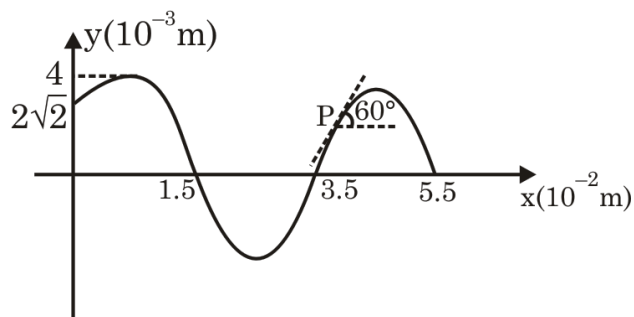
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Three concentric conducting spherical shells have radius R , $2R$ & $3R$ are shown. Middle shell is earthed and inner most shell is connected to outer most shell by conducting wire. Space between 2^{nd} and 3^{rd} shell is filled by dielectric of dielectric constant $K = 2$ and $Q = 17\mu\text{C}$ charge is given to 3^{rd} shell then :-

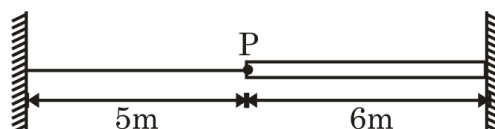


- (A) Capacitance of system is $44\pi\epsilon_0 R$ (B) Capacitance of system is $68\pi\epsilon_0 R$
 (C) Charge on 2^{nd} shell is $-14\mu\text{C}$. (D) Charge on 2^{nd} shell is $-17\mu\text{C}$.

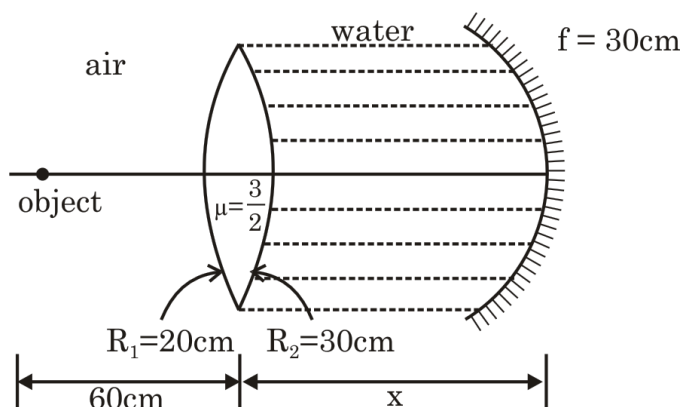
8. The figure shows a snap shot of a vibrating string of linear mass density 0.1 kg/m at $t = 0$. The particle P is observed moving up with velocity $20\sqrt{3} \text{ cm/s}$. The tangent at P makes an angle 60° with x-axis then



- (A) The wave equation is given by $y = 4 \text{ mm} \sin \left[10\pi t + 50\pi x + \frac{\pi}{4} \right]$.
- (B) The wave equation is given by $y = 4 \text{ mm} \cos \left[10\pi t - 50\pi x + \frac{\pi}{4} \right]$.
- (C) Rate of flow of energy is towards left and it's value at point P is $12 \times 10^{-5} \text{ J}$
- (D) Tension in string is $4 \times 10^{-3} \text{ N}$.
9. Two string having linear mass density 0.1 kg/m and 0.4 kg/m are connected as shown and pulled by force 10 N . Select correct alternative :-

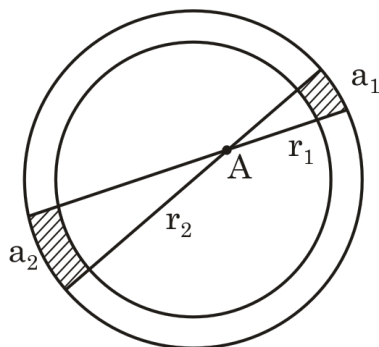


- (A) Fundamental frequency of system is 10 Hz , if antinode is formed at point P.
- (B) Fundamental frequency of system is 5 Hz , if node is formed at point P.
- (C) 1^{st} overtone has frequency 30 Hz , if antinode is formed at point P.
- (D) 1^{st} overtone has frequency 10 Hz , if node is formed at point P.
10. In given situation find possible values of distance x between lens and mirror so that final image forms at object itself. Refractive index of water is $4/3$.

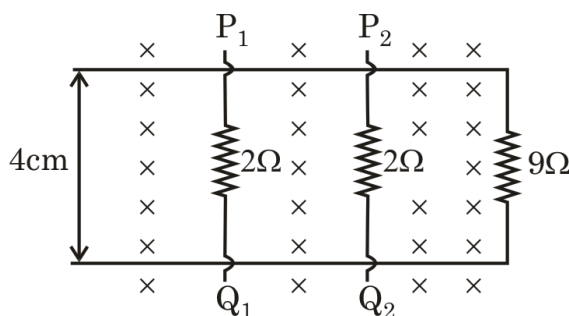


- (A) 96 cm (B) 36 cm (C) 66 cm (D) 156 cm

11. A wire having a uniform linear charge density λ , is bent in the form of ring of radius R . Point A as shown in the figure, is in the plane of ring but not at centre. Two elements of the ring of length a_1 and a_2 subtend very small same angle at the point A. They are at distance r_1 and r_2 from the point A respectively.



- (A) The ratio of charge of elements a_1 and a_2 is $\frac{r_1}{r_2}$.
- (B) The element a_1 produced greater magnitude of electric field at A than a_2 .
- (C) The element a_1 and a_2 produced same potential at A.
- (D) The direction of net electric field at A is towards element a_2 .
12. In given figure the wires P_1Q_1 and P_2Q_2 are made to slide on the rails with same speed 5 cm/s. In this region a uniform magnetic field 1T exists inside plane of paper. The current in 9Ω resistor is :



- (A) Zero if both wires slide towards left.
- (B) Zero if both wires slides in opposite direction.
- (C) 0.2 mA if both wire move towards left.
- (D) 0.2 mA if both wire move in opposite direction.

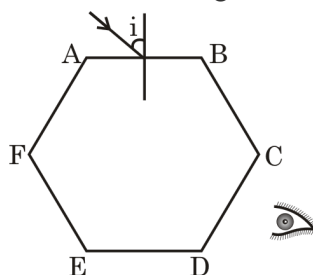
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

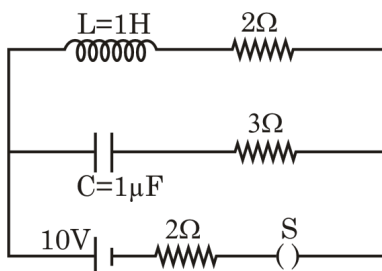
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A water jet shoot water with speed 5 m/s from an opening of area 15 cm^2 in upward direction. A plate is balanced at height 80 cm above jet by force applied by water. After colliding the plate water goes horizontally in every possible direction. Assuming water to be ideal fluid find angular frequency (in rad/s) of small oscillation of plate in vertical direction. (Take $g = 10 \text{ m/s}^2$, density of water = 10^3 kg/m^3)
2. A current $I = 10\text{A}$ flows in a long thin walled cylinder of radius 1cm. Then pressure on walls due to magnetic force is (in Pa)
3. A particle is projected from point, that is at a distance $4R$ from centre of the earth, with speed $50\sqrt{2} \text{ m/s}$, in a direction making angle 30° with vertical, away from the earth. Find x if particle passes grazing the surface of the earth use $\frac{GM}{R} = 6.4 \times 10^7 \text{ m}^2/\text{s}^2$, where R is radius of earth and M is mass of the earth.
4. Position of a particle moving along x axis is given by $x = A \cos\left(\frac{\pi}{180}t\right)$. Find time (in sec) in which particle will travel a distance $\frac{12A}{5}$.
5. A transparent hexagonal body ABCDEF is shown find minimum value of refractive index of body so that ray incidenting at surface AB do not reach observer through surface CD.



6. At $t = 0$, key S is closed. If I_1 represent current through cell just after key is closed while I_2 represent current long after key is closed then $I_1 + I_2$ will be (in A).



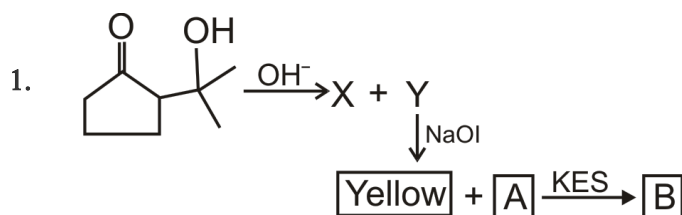
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

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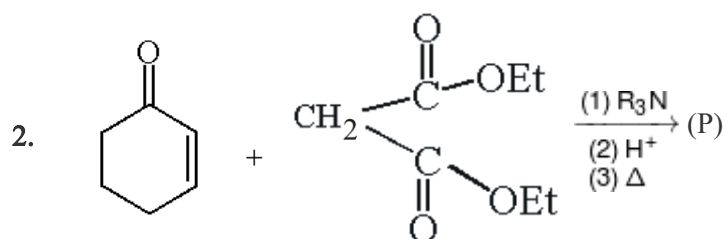
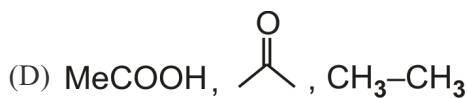
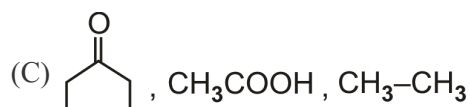
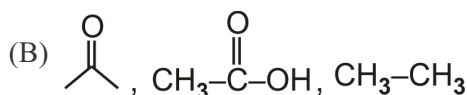
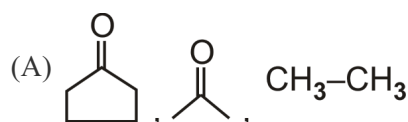
Full Marks : +3 If **ONLY** the correct option is chosen.

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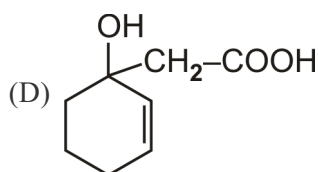
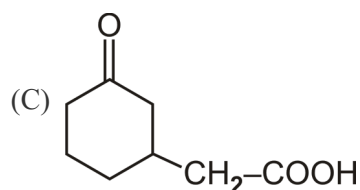
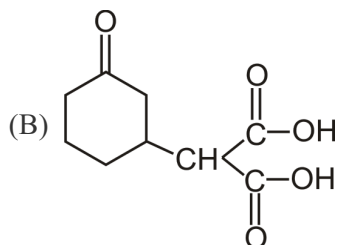
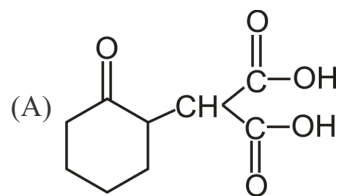
Negative Marks : -1 In all other cases

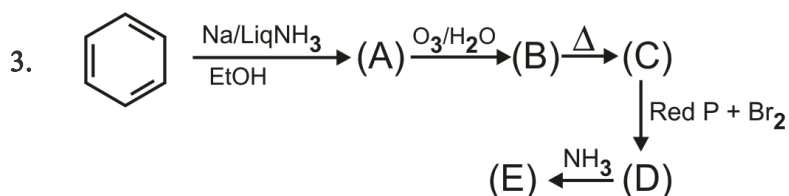


in above reaction Y, A, B respectively is :-

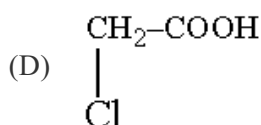
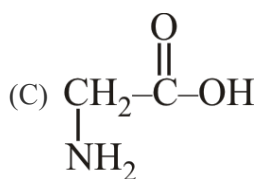
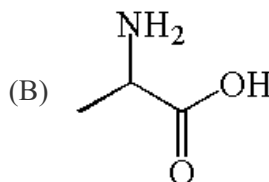
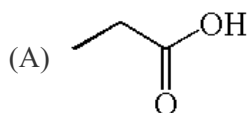


product P is ?





Product E is ?



4. Which of the following is **INCORRECT**?

(A) H_2S does not give test with sodium nitroprusside

(B) HgCl_2 does not give chromyl chloride

(C) Mg^{2+} can not be detected by flame test

(D) Through blue glass brick red flame is observe for Ca^{+2} in flame test

5. Which of the following set is correct?

(A) L.D. Process $\rightarrow$ Used for : Manufacturing of steel

(B) Hall's Heroult Process $\rightarrow$ Used for : Conversion of bauxite ore into pure alumina

(C) Poling Process $\rightarrow$ Used for : Purification of impure Cu

(D) Liquation Process $\rightarrow$ Used for : Purification of impure Pb

6. Which of the following having density less than Na?

(A) K

(B) Rb

(C) Cs

(D) None of these

SECTION-I (ii) : (Maximum Marks: 24)

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Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

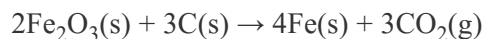
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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. The standard boiling point of $A(l)$ is 400K. The enthalpy of vapourisation $A(l)$ at 400 K and 1 bar is 20 kcal/ mol. Which of the following information (s) is/are correct regarding the process $A(l, 400K, 1 \text{ bar}) \longrightarrow A(g, 400K, 0.5 \text{ bar})$
[Given : $C_{p,m}$ values (in cal / K-mol) : $A(g) = 5$; $A(l) = 10$; $\ln 2 = 0.7$, $\ln 5 = 1.6$, Assume ideal behaviour of $A(g)$]
- (A) $\Delta H_m = + 20 \text{ kcal/ mol}$ (B) $\Delta S_m = + 45.1 \text{ cal/ K-mol}$
(C) $\Delta G_m = - 560 \text{ cal/ mol}$ (D) $\Delta U_m = 20 \text{ kcal/mole}$
8. Select the correct statements
- (A) Catalyst can change the percentage yield of a reversible reaction
(B) Molecularity and order are same for the elementary reactions
(C) The catalyst forms an activated complex of lower potential energy, with the reactants by which more number of molecules are able to cross the barrier per unit of time.
(D) For exothermic reaction rate constant decreases with increasing temperature

9. Consider the following reaction :



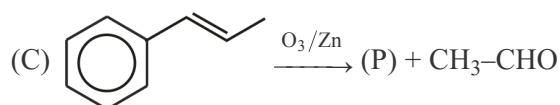
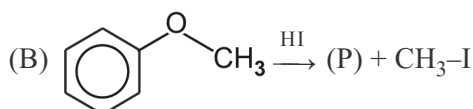
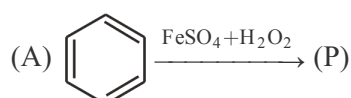
ΔH_f° of Fe_2O_3 & CO_2 are -820 kJ/mol and -390 kJ/mol respectively.

The reaction is -

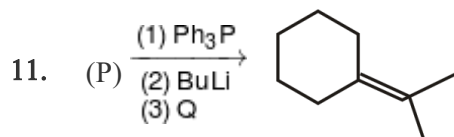
- (A) Endothermic (B) Exothermic
(C) Spontaneous at high temperature (D) Spontaneous at low temperature

10. $\text{P} + \text{HCHO} \xrightarrow{\text{OH}^- / \text{Polymerisation}} \text{Bakelite}$

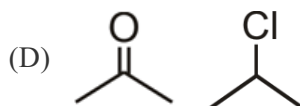
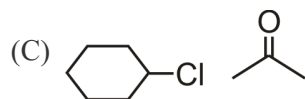
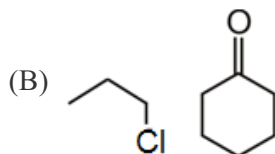
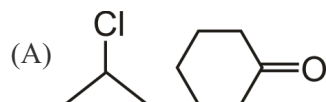
Find the reaction in which P form as product.



(D) None

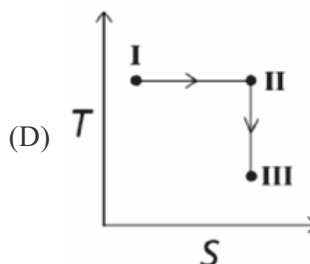
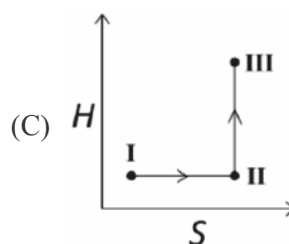
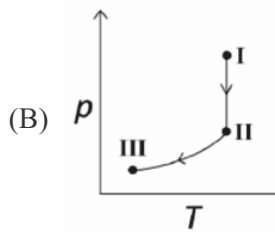
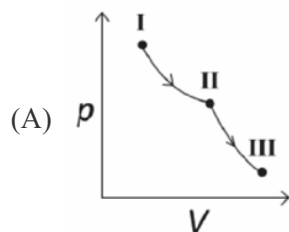


Possible P and Q is/are :-



12. An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state II to state III. The correct plot(s) representing the changes from state I to state III is(are)

(p: pressure, V: volume, T: temperature, H: enthalpy, S: entropy)



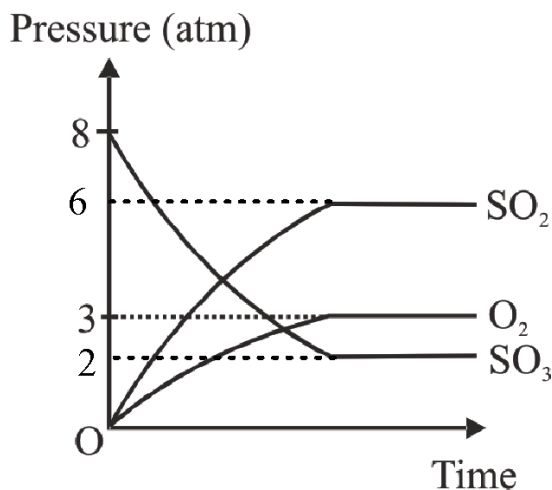
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

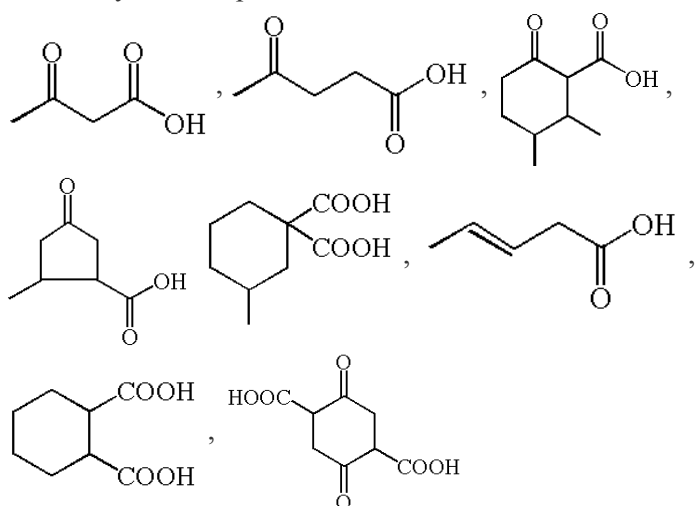
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. 1 molal aqueous solution of an electrolyte AB_2 is 80% ionised in to A^{+2} and B^{-1} ions. The boiling point of the solution (in Kelvin) at 1 atm is : ($K_b(H_2O) = 0.5 \text{ K kg mol}^{-1}$, $0 \text{ K} = -273^\circ\text{C}$).
2. For the dissociation of SO_3 into SO_2 and O_2 , at a given temperature, the equilibrium constant $K_p = x \text{ atm}^2$, then find x. The variation of the partial pressure of the three gases with time is as shown :



3. Examine the structure of following compounds, and find out number of compounds that will undergoes decarboxylation in presence of heat.



4. Calculate the magnetic moment of $[Co(H_2O)_6]^{3+}$ in B.M.
5. What is the average oxidation of 'S' in product in I_2 reacts with hypo solution.
6. The crystal field stabilizing energy of $[IrF_6]^{3-}$ is $-x\Delta_0 + yP$.
find the sum of the magnitude of $x + y$ is

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $A = \begin{bmatrix} x^2 \\ 1 \\ 2x \end{bmatrix}$, $B = \begin{bmatrix} a & b & c \end{bmatrix}$ and

$C = \begin{bmatrix} 3x^2 & 5x^2 & 2x \\ 2x & 4 & 3x \\ 5x & 7 & 6x \end{bmatrix}$, given

$\text{tr}(AB) = \text{tr}(C) \forall x \in \mathbb{R}$, where $\text{tr}(A)$ denotes trace of A , then the value of $\int_0^{\infty} \frac{\ln x}{ax^2 + bx + c} dx$

- (A) 0 (B) 1
(C) -3 (D) 3

2. The integral curve through origin satisfying the differential equation $y^2(3x^9 + y) \frac{dy}{dx} = x^3(1 - 9x^5y^3)$ is

- (A) $3(x^3y)^2 + y^3 - x^3 = 0$
(B) $4(x^3y)^3 + y^4 - x^4 = 0$
(C) $4(x^3y)^3 - y^4 + x^4 = 0$
(D) $3(x^3y)^2 - y^3 + x^3 = 0$

3. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & -2 & 3 \end{bmatrix}$ & $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then if $A^7 - 4A^6 + 6A^5 = \alpha A^2 + \beta A + \gamma I$ then (α, β, γ) is :

- (A) $(5, -11, 7)$
 (B) $(-21, 115, -91)$
 (C) $(-13, 44, -28)$
 (D) None of these

4. If the solution of the differential equation $\frac{xdx - ydy}{xdy - ydx} = \sqrt{\frac{1 + x^2 - y^2}{x^2 - y^2}}$ be

$\sqrt{f(x, y)} + \sqrt{1 + f(x, y)} = c \left(\frac{x + y}{\sqrt{f(x, y)}} \right)$, then $f(x, y)$ is :

- (A) $x^2 + y^2$
 (B) $1 + x^2 - y^2$
 (C) $x^2 - y^2$
 (D) $\frac{x^2 - y^2}{x^2 + y^2}$

5. $\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + n + 1} - \left[\sqrt{n^2 + n + 1} \right] \right)$ ($n \in \mathbb{N}$) (where $[.]$ denotes the greatest integer function) is

- (A) 0
 (B) $1/2$
 (C) $2/3$
 (D) $1/4$

6. A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it, such that there will be no complete pair is :

- (A) 1920
 (B) 200
 (C) 110
 (D) 80

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. If the solutions of the equation $x^3 + 1 = 2\sqrt[3]{2x-1}$ are $\left\{a, \frac{b \pm \sqrt{c}}{d}\right\}$, ($a, b \in \mathbb{I}$ & c, d are prime) then which of the following is(are) correct ?

(A) $a + b^3 + c - d = 3$ (B) $a + b = 1$

(C) $c + d = 7$ (D) $c - d = 3$

8. A function f is defined by $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good?

(A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$

(B) Maximum value of $f(x)$ is π

(C) There exists atleast one $c \in (0, 2\pi)$ such that $f'(c) = 0$

(D) Minimum value of $f(x)$ is $-\frac{\pi}{2}$

9. Given a function 'g' which has a derivative $g'(0) = 2$ and $g(x+y) = e^y g(x) + e^x g(y)$ for all x and y , then:
- (A) $g(x)$ is increasing for all $x \in [-1, \infty)$ (B) Range of $g(x)$ is $\left[-\frac{2}{e}, \infty\right)$
- (C) $g''(x) > 0 \forall x \in \mathbb{R}$ (D) $\lim_{x \rightarrow 0} \frac{g(x)}{x} = 1$
10. If y_1, y_2 are the solutions of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, then :
- (A) $y = y_1 + c(y_1 - y_2)$ is the general solution of equation
- (B) $y = y_1 + c(y_1 + y_2)$ is the general solution of equation
- (C) $\alpha y_1 + \beta y_2$ is a solution if $\alpha + \beta = 1$
- (D) $\alpha y_1 - \beta y_2$ is a solution if $\alpha - \beta = 1$
11. Area of region bounded between the curves $y = \frac{1}{1+x^2}$ and $y = 1 - \frac{1}{1+x^2}$ between its points of intersection is
- (A) $\pi - 2$ (B) an irrational number
- (C) an integer (D) 4
12. The combinatorial coefficient ${}^{n-1}C_p$ denotes
- (A) The number of ways in which n things of which p are alike and rest different can be arranged in a circle
- (B) The number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded
- (C) Number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty, and each box can hold any number of balls
- (D) The number of ways in which $(n-2)$ white balls and p black balls can be arranged in a line if black balls are separated, balls are alike except for the colour

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Let solution of equation $|x^2 - 3x - 4| + |x^2 + 2x - 3| = |5x + 1|$ is $x \in [a, b] \cup [c, d]$, then value of $|a| + |b| + |c| + |d|$ is
- Consider the matrix A, B, C, D with order $2 \times 3, 3 \times 4, 4 \times 4, 4 \times 2$ respectively. Let $x = (\alpha A B \gamma C^2 D)^3$, where α & γ are scalars. Let $|x| = k |A B C^2 D|^3$ & if $k = (\alpha \gamma)^\lambda$, then λ is (where $|P|$ denotes determinant value of matrix P)
- If $A = \frac{d \left(\sin^{-1} \left(\frac{x}{\sqrt{4+x^2}} \right) \right)}{d \left(\tan^{-1} \frac{x}{2} \right)} \Bigg|_{x=0}$ and $B = \frac{d \left(\sin^{-1} x \right)}{d \left(\tan^{-1} \frac{\sqrt{1-x^2}}{x} \right)} \Bigg|_{x=\frac{1}{2}}$, then the value of $(2A + 3B^2)$ is
- If $I_1 = \int_0^1 \frac{dx}{e^x (1+x)}$ and $I_2 = \int_0^{\pi/4} \frac{e^{\tan^2 \theta}}{(2 - \tan^2 \theta)} \frac{\sin \theta d\theta}{\cos^3 \theta}$, then $\frac{I_1}{I_2}$ is _____.
- The area bounded by the curves $y = xe^x$, $y = xe^{-x}$ and the lines $x = 0$ and $x = 1$ is (in square units) :
- Let $A = [a_{ij}]_{3 \times 3}$, $B = [b_{ij}]_{3 \times 3}$ where $b_{ij} = 3^{i-j} a_{ij}$, $C = [c_{ij}]_{3 \times 3}$, where $c_{ij} = 4^{i-j} b_{ij}$ be any three matrices. If $|A| = 4$, then $|B| - |C|$ ($|X|$ denotes determinant of matrix X)

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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